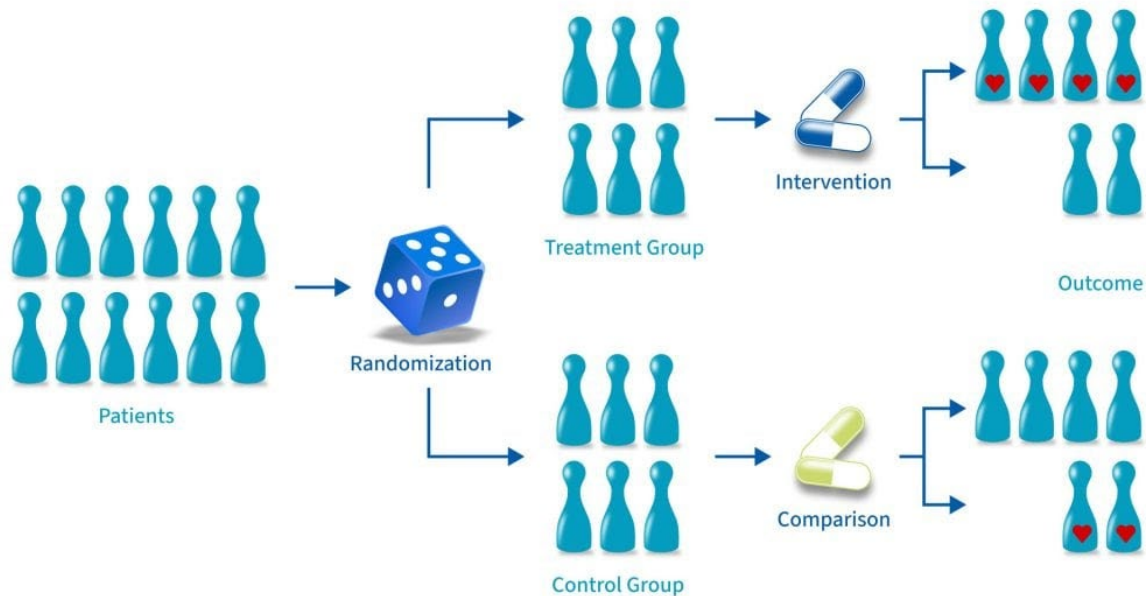


Randomized Controlled Trial (RCT): What are the key characteristics? by Carsten Bergenholtz (ChatGPT-4 used to check for errors), 10.01.25

A RCT – also called an experiment - is a particular kind of research design. Rather than just looking at data that might be available, one collects data in a particular way, with the aim of being very certain about the causality one can identify. The key elements of a RCT are visualized in the following image.

Randomized Controlled Trial



<https://www.simplypsychology.org/randomized-controlled-trial.html>

At the left we start with a group of **participant** (patients) that are then **randomly divided** into two groups, who each receive a different (or no) **intervention** (also called treatment). Now we can investigate if the **outcomes are different** for the patients in the two groups. In this way, the outcome of the control group shows the counterfactual: The outcome that would have occurred if the treatment had not been in place.

The aim of this text is to explain the core elements of an RCT, and why it constitutes a particular research design; some categorize it as the gold standard, for establishing causality. First, I explain how a RCT allows us to compare an outcome with the counterfactual, then I present the key elements of an experiment, how to assess the validity of RCTs and some ethical considerations. Note that we in this text focus on people as being part of experiments, but experiments can also involve teams, firms, regions, webpages and other kinds of units of analysis.

Causality: RCT allows a comparison of an outcome with the counterfactual

In research, we are often interested in if an independent variable (e.g. getting access to chatbots while doing an exam) impacts a dependent variable (e.g. performance in an exam). In a RCT, we manipulate the **independent variable (IV)** to determine if it impacts the **dependent variable (DV)**.

This manipulation is the treatment, and we then compare a treatment group (or multiple treatment groups) with a control group, which then constitutes the counterfactual: what would happen without the manipulation (e.g. getting access to a chatbot). The aim is to ensure that everything is kept constant between the groups, except for the treatment.

Examples of Treatments and Control Groups

1. *Aspirin on headache*: One group receives aspirin while the other group receives a placebo. By comparing the two, we can determine if aspirin has a significant effect on headache relief.
2. *Chatbot usage impact on exam performance*: One group has access to chatbots for studying, while the control group does not. Differences in performance could indicate the impact of chatbot usage.
3. *Size of incentives and performance*: One group receives a small (monetary) incentive, while another group receives a larger incentive. The difference in their performance levels can reveal the effect of incentive size on performance (this one is used as an example in Bergenholtz 2025b, curriculum).

However, not all comparisons are valid. We cannot simply compare individuals who decided to get access to chatbots to those who did not — this could introduce selection bias or other **confounding** factors. For example, maybe those that want to use chatbots are different from those that do not. Instead, an RCT ensures that the only difference between the two groups is who receives the treatment, achieved by **random assignment**.

Randomization

In the context of a RCT, **internal validity** refers to the degree to which we can be confident that the treatment caused the outcome, without interference from other factors. More generally, internal validity means how confident we can be that a particular research study establishes something that is true.

In essence, by **randomizing** people into **treatment** and **control** groups, we ensure that the two groups are *as similar as possible* - and the only difference is the manipulation. Imagine a lab-experiment where we randomize people into two groups; they all have to do the same tasks in the same environment, just with one group having access to ChatGPT-4, and the other not having access to such chatbots. In this way, the randomization ensures that the control group constitutes a counterfactual.

Consider the following numbers and coin flip example. Just like a coin flip introduces randomness, assigning participants to different groups in an RCT follows the same principle, ensuring that each participant has an equal chance of being in either the treatment or control group. However, if we flip a coin 4 times we might get 3 tails and 1 head. These two outcomes of coins are not equal or even nearly equal. Now, assume you flip the coin 150 times (or more). The outcome is more likely to be almost equal; maybe 77 head and 73 tails. Thus, when you randomize a non-small group of people into different groups, we can more certain that people in each group are, in effect, identical (same gender distribution, same age, same abilities etc.). The following table illustrates the point; note the sample size is 100:

	Group	Age	Experience (Years)	Motivation Score	Female	Male
1	Control	25.0	3.8	7.4	29	21
2	Treatment	24.1	3.7	7.5	26	24

(Table and randomization generated by ChatGPT-4).

The distribution is not perfect; there are some differences, but they are small and we can assume that they do not matter. We often do such a numerical comparison of key characteristics of participants in an experiment, to verify randomization has worked properly.

In Isager's (2023) terms, this randomization approach guards against **selection bias** and **confounding variables**, ensuring clear causal direction and high internal validity. People have not self-selected into the experiment (creating the risk that people who like chatbots end up in the chatbot condition) and people are almost identical, ensuring that no confounding variables disturbs our comparison of treatment vs. control group.

Treatment allocation (allows comparison to control group)

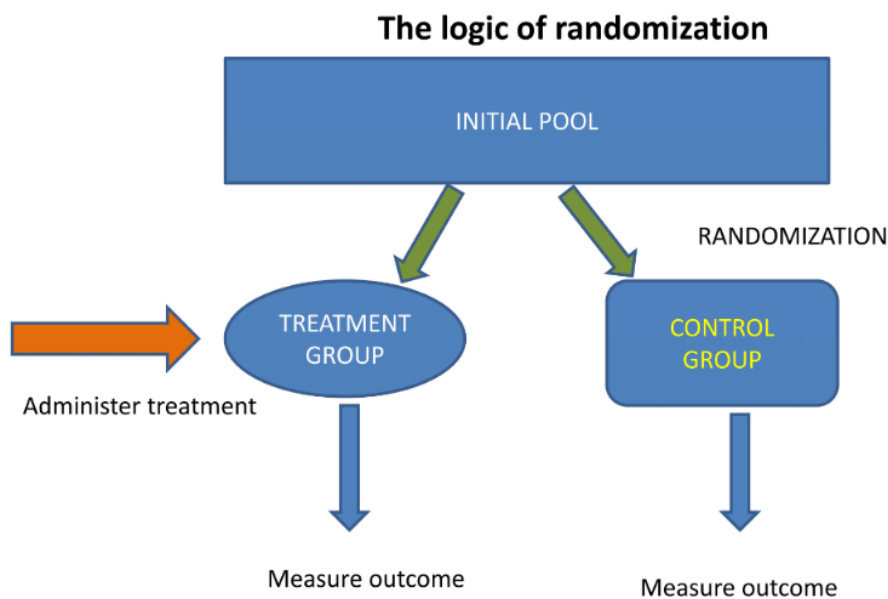
Once participants are randomized into their respective groups, the treatment (the variable being tested) is applied to the treatment group, while the control group does not receive the treatment. In some cases, the control group might receive a placebo, depending on the study design. For example, consider the chatbot example. Participants in the treatment group are given access to the chatbot, while students in the control group continue studying without it. Everything else, such as the study environment and materials, remains the same for both groups. This step ensures that the treatment is the only variable that differs between the groups, making it easier to determine whether the treatment (chatbot access) has a causal impact on the outcome. In essence, the control group is the counterfactual – what would have happened without the treatment. If the randomization process had not worked, and the people in the group were not really identical, then we would risk that it is not the treatment that leads to different outcomes, but the people in the respective groups.

Measuring outcome and analysis of RCT

In order to enable a comparison of treatment vs. control condition, one should always measure some kind of outcome (e.g. performance on a task, with or without having access to chatbots). This is done when the task is completed. This data is then analysed quantitatively, determining whether one group's outcome is statistically significantly different from the other's. This approach provides an **average effect**, across all participants in the group. For example, if participants with chatbot access perform significantly better, we conclude the treatment had an impact, on average.

To summarize

A Randomized Controlled Trial (RCT) is built on **three key elements: randomization, treatment, and a control group (or controlled environment)**. Randomization ensures that participants are assigned to groups by chance, eliminating bias and balancing out confounding factors. The treatment is the variable being tested, applied only to the treatment group, while the control group receives no treatment, allowing researchers to statistically compare outcomes and establish the counterfactual analysis and result.



If outcome for treatment group $\neq$ outcome for control, the administered treatment caused it

(figure borrowed from Phanish Puranam)

Bergenholtz 2025b presents a more detailed illustration of what an actual RCT looks like in practice, including some insights on typical challenges as well as an illustration of how to assess the external validity, cf. next section.

External validity criteria

The prior sections have explained what a RCT is and why it has high internal validity (that is why we can be confident that we have identified a clear causal effect in the particular sample).

To ensure findings are applicable beyond the specific context of the study, the external validity should be considered. **External validity** involves assessing if results can be transferred to or generalized to other situations (e.g. other empirical settings).

Four core criteria must be put in play to evaluate external validity effectively:

1. **Interaction of selection and treatment:** What group of people were part of the experiment, and can the results apply to different groups of people?
2. **Interaction of setting and treatment:** In what setting did the experiment take place. E.g., what task should the participants complete in the experiment?
3. **Interaction of history and treatments:** Society changes. Would the results hold true at a different time? E.g., might results from 1960 hold true for 2025?
4. **Hawthorne effect:** Might the results be shaped by people knowing they were part of an experiment? Recall insights from Organizational Behavior, and that people might change behavior if they know they are being observed.

(list of criteria based on Bell et al. 2022, p. 54-55).

Assessing the above four elements is key for evaluating the external validity of an RCT. Hence if you are asked to assess external validity, keep these in mind.

Additionally, and relatedly, we can consider the **ecological validity** of an experiment. While external validity concerns if an experiment can generalize to other people, settings and times, ecological validity specifically is specifically about how "naturalistic" the experiment is. E.g., a lab-experiment is often somewhat artificial, and might not resemble how the real world works. This would mean that it has low ecological validity. Consider the differences between how one might give people different incentives to complete tasks in a lab-experiment (potentially lower ecological validity), vs. getting a raise in your actual job (high ecological validity)

Lab vs. Field Experiments

We can also consider different types of experiments. Lab and field experiments are quite different:

Lab Experiments: Conducted in **artificial** environments (e.g. the COBE Lab at BSS) where it is easier to control what is going on. This ensures high internal validity. However, they may lack ecological validity — the extent to which the conditions reflect real-world settings.

Field Experiments: Conducted in **natural** settings (e.g. in a company), offering higher ecological validity but less control over external and potentially confounding variables. Since the study takes place in natural settings, the outcomes are likely more realistic and potentially more generalizable. However, it is also more difficult to control all confounding variables, which can lead to lower internal validity.

Thus, lab experiments are carefully controlled, making it easier to isolate the treatment effect (higher internal validity), but they may not represent real-world conditions (lower ecological validity). In contrast, field experiments are conducted in real settings, making them more applicable to real-world situations (higher external validity), though harder to control for all variables. For example, manipulating the impact of incentives on employee performance can be done in a lab setting with artificial tasks (lab-experiment) or in an actual workplace (field-experiment). The lab ensures control (everything is the same for both groups), while the field reflects real-world complexities (yet, how certain can we be that people in different groups did not communicate, for example?).

Quasi-Experiments

Moving beyond the pure experiment, we can consider research designs that are almost as good as an actual experiment; quasi-experiments. These resemble RCTs but lack full internal validity, often due to the absence of randomization. They are "studies that have certain characteristics of experimental designs but do not fulfil all the internal validity requirements" (Bell et al. 2022: p. 57). See Bergenholtz 2025c (curriculum), for a more in-depth presentation of what quasi-experiments are (not), including a few examples of quasi-experiments.

Limitations and ethical concerns of RCTs

Practical Challenges: Conducting RCTs in business settings presents several **logistical** and **methodological challenges** that can limit their effectiveness. One common issue is the risk of contamination between treatment and control groups. In a workplace, employees often communicate across groups, sharing information or behaviors that can affect the experiment's outcomes. For example, if one group is given access to a new software tool, they might discuss its benefits with colleagues in the control group, thereby reducing the contrast between the two groups and weakening the study's validity.

Moreover, business environments are often **complex** and influenced by many external factors, such as economic conditions, company culture, and market dynamics. These factors can make it difficult to isolate the effect of the treatment being tested, reducing the internal validity of the RCT (more on this in a later lecture, on complexity). Some business variables, such as organizational culture or leadership style, are also difficult to manipulate experimentally. For instance, cultural changes in an organization are slow-moving and cannot be easily randomized or controlled over short time periods. Furthermore, businesses may resist the disruption caused by RCTs, as implementing experimental conditions often requires changing workflows or resources, which can be difficult in a real world, ever-changing environment.

These challenges highlight that while RCTs are powerful for establishing causality, they are not always suitable for every business research question. The complex and dynamic nature of business environments often means that we need to supplement RCTs with other research methods, such as longitudinal studies, qualitative approaches, or observational data, to capture a fuller picture of the factors at play.

Ethics: While RCTs are a powerful tool for establishing causality, they can raise significant **ethical concerns** in business settings. One key issue is the potential harm or disadvantage to participants in the control group, particularly in organizational experiments. For instance, testing new employee policies—such as a more flexible work schedule—might benefit the treatment group while leaving the control group at a disadvantage. Not being in the treatment (or control) group could lead to job dissatisfaction. Moreover, withholding beneficial interventions, like access to leadership development programs, could be seen as unethical if the organization already suspects these programs to be helpful. In such cases, researchers must carefully consider the ethical implications of not letting everyone get access to potential benefits.

Additionally, employees in a business setting are often not in the same position as voluntary participants in psychological or medical trials. They may feel forced to participate in the experiment due to workplace pressures or fear of reprisal, which raises concerns about informed consent. Ensuring that participation is truly voluntary and that participants fully understand the study's risks and benefits is essential. Researchers need to be transparent about the purpose of the RCT and guarantee that no employee will face negative consequences for their role in the study, whether they are in the treatment or control group.

References

Bell, Emma, Alan Bryman, and Bill Harley (2022). *Business research methods*. Oxford University Press, 2022

Isager (2023): From curriculum – see

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